

ACADEMIC ESSAY

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The Second Epidemic

Applying epidemiological models to health misinformation in Bangladesh

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Abstract

The World Health Organization's description of COVID-19 as an "infodemic" alongside a pandemic is usually read as rhetoric. This paper takes it as a methodological proposition and asks how far the core apparatus of epidemiology — the agent–host–environment triangle, the basic reproduction number and the herd-immunity threshold, surveillance, and the three levels of prevention — can be applied to health misinformation as a genuine analytic framework rather than a figure of speech. Using documented harms from the COVID-19 infodemic and cases from Bangladesh, it argues that misinformation behaves enough like a communicable disease for epidemiological models to yield real purchase: transmission can be modelled, an effective reproduction number can be lowered, and a form of herd immunity can be built through "prebunking." It then marks the point at which the analogy fails — misinformation is not a value-neutral pathogen, and treating it as one can license its own iatrogenic harm, the over-broad censorship that mistakes dissent for disease. The conclusion argues for infodemiology as a standing capacity within Bangladeshi public health, integrated with, not substituted for, the surveillance of biological disease.

1. The metaphor made literal

In February 2020, weeks before most governments grasped the scale of what was coming, the Director-General of the World Health Organization told the Munich Security Conference that "we're not just fighting an epidemic; we're fighting an infodemic" (World Health Organization, 2020). The line was widely quoted as a warning about the noise surrounding the new coronavirus. It was more than that. It was a claim that false information had itself become a public-health problem — an object with its own morbidity and mortality, worthy of surveillance and control by the same institutions that track a virus.

The claim can be tested, because the COVID-19 infodemic left a body count. In the most cited single study of its harms, Islam and colleagues (2020) analysed 2,311 reports of rumours, stigma, and conspiracy theories circulating in 25 languages across 87 countries in the first months of 2020, of which the great majority were rumours. Acting on one of them — the belief that highly concentrated alcohol disinfects the body of the virus — roughly 800 people died of methanol poisoning, some 5,876 were hospitalised, and 60 developed complete blindness (Islam et al., 2020). These are not the metaphorical casualties of a "toxic" information environment. They are

deaths, admissions, and disabilities with a proximate cause that a coroner could name, and their cause was a sentence.

This paper takes the infodemic seriously as an epidemiological object and asks a disciplined question: how much of the actual machinery of epidemiology — not its vocabulary but its models — can be applied to health misinformation, and where does the application break down? The argument proceeds in four steps. It first traces how the analogy became a method, in the sub-discipline of infodemiology. It then applies epidemiology's foundational model, the agent–host–environment triangle, to misinformation, and its central quantitative tools, the reproduction number and the herd-immunity threshold, to the spread of a false claim. It then reorganises the response around epidemiology's three levels of prevention. Finally, and most importantly, it marks the limits of the analogy, because a model applied past its domain does its own damage — and in the governance of information that damage has a specific and familiar shape. Throughout, the cases are drawn from Bangladesh, where the infodemic is not an abstraction: rumours here have moved crowds to kill.

2. From metaphor to method: infodemiology

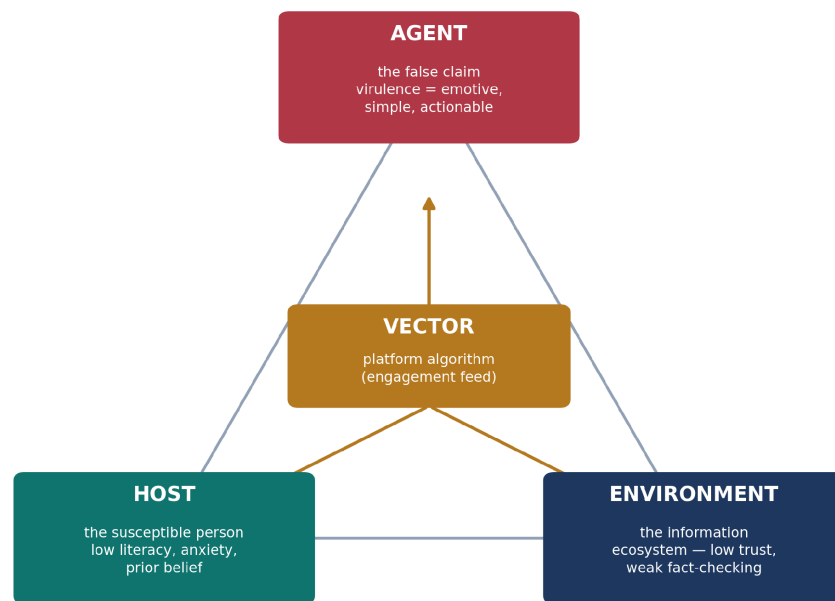
The word "infodemic" predates COVID-19, and so does the discipline built to study it. Gunther Eysenbach coined "infodemiology" — the epidemiology of information — in the early 2000s and developed it into a research programme for analysing the distribution and determinants of information in an electronic medium, with the express purpose of informing public health (Eysenbach, 2009). Its surveillance arm, "infoveillance," proposed to track health information the way public health tracks disease: monitoring search queries, social-media posts, and news to detect emerging problems in real time. For two decades this was a niche field. The pandemic made it central, and in 2020 the WHO operationalised it, convening the first infodemiology conference and publishing a framework for "infodemic management" built explicitly on public-health logic (Eysenbach, 2020).

What infodemiology asserts, and what this paper tests, is that the analytic tools epidemiology developed for pathogens are not merely evocative when turned on misinformation but genuinely productive — that they generate predictions, identify leverage points, and structure interventions. The claim is plausible on its face because the two phenomena share a deep structure. Both

involve something transmissible passing between hosts through identifiable channels; both show exponential growth in a susceptible population; both exhibit superspreading, clustering, and waves; and in both the decisive quantity is not the number already affected but the rate at which each case produces the next. The sections that follow put that shared structure to work.

3. The epidemiologic triangle of the infodemic

The foundational model of communicable-disease epidemiology is the epidemiologic triangle: disease arises not from a pathogen alone but from the interaction of an agent, a susceptible host, and a permissive environment, often with a vector that carries the agent from source to host (Centers for Disease Control and Prevention, 2012). The model's discipline is that it forbids single-cause explanations. Malaria is not caused by the *Plasmodium* parasite alone; it requires the parasite, a non-immune human, the *Anopheles* mosquito, and the standing water and warmth that let the mosquito breed. Remove any vertex and transmission collapses. Applied to health misinformation, the same four-part analysis both organises the phenomenon and, more usefully, exposes where it can be interrupted (Figure 1).



Transmission occurs when a virulent claim meets a susceptible host through a competent vector in a permissive environment.

Figure 1. The epidemiologic triangle applied to health misinformation. Transmission requires all four elements; removing any one interrupts it.

The *agent* is the false or misleading claim, and like a pathogen it has a virulence that varies by strain. The claims that spread are not random; they are selected for transmissibility. A claim that is emotionally charged, cognitively simple, and immediately actionable — "alcohol kills the virus," "the vaccine contains a microchip," "child-kidnappers are taking heads for the new bridge" — out-competes a claim that is complex, qualified, and undramatic, in exactly the way a more transmissible variant out-competes a less transmissible one. The *host* is the individual who receives the claim, and hosts differ in susceptibility. Low health literacy and low media literacy raise it; so do anxiety and fear, which is why infodemics peak precisely when populations are most frightened; so does congruence with prior belief, the cognitive analogue of a receptor the pathogen already fits. The *environment* is the information ecosystem: the level of institutional trust, the strength or weakness of independent fact-checking, the density of press freedom, and the acuteness of the surrounding crisis. A society with low trust in its health authorities and thin fact-checking capacity is, epidemiologically, a body with a compromised immune system — the same agent produces far more disease in it.

The fourth vertex is where the contemporary infodemic departs from its historical predecessors, and where this analysis meets the wider study of platforms. The *vector* is the mechanism of transmission, and today that mechanism is the algorithmically curated feed. A vector in epidemiology is judged by its competence: how efficiently it acquires the agent, sustains it, and delivers it to new hosts. By that standard the engagement-optimised recommender system is an extraordinarily competent vector, because the very features it is built to maximise — emotional arousal, controversy, shareability — are the features that define a virulent claim. The landmark study by Vosoughi, Roy, and Aral (2018), analysing the diffusion of verified true and false stories across Twitter over more than a decade, found that falsehood spread significantly farther, faster, deeper, and more broadly than the truth in every category of information, and that human beings, not automated accounts, drove the difference — because false news was more novel and provoked stronger reactions. The vector, in other words, does not spread all agents equally; it preferentially amplifies the virulent ones. This is a mosquito that prefers to carry the deadliest strain.

Bangladesh supplies the triangle with grim illustrations at every vertex. In mid-2019 a rumour that the construction of the Padma Bridge required human heads fused with an older panic about

child-kidnappers; the agent was maximally virulent — a claim about children, about sacrifice, about a national megaproject, demanding immediate protective action. The hosts were a population primed by fear and low trust; the environment was one of weak correction and high rumour density; and the vector was Facebook, on which the claim and lists of supposed kidnappers circulated faster than any denial. At least eight people were beaten to death by mobs, among them Taslima Begum Renu, a mother lynched outside a school in Dhaka after being accused, on no evidence, of being a kidnapper (Al Jazeera, 2019; BenarNews, 2019). The infodemic is not, in Bangladesh, a Northern anxiety about vaccine hesitancy. It has produced corpses.

4. Transmission dynamics: reproduction, superspreading, and the misinformation curve

If the triangle describes the conditions for transmission, the reproduction number describes its dynamics. The basic reproduction number, R_0 , is the average number of secondary cases produced by one case in a fully susceptible population; it is the single most important quantity in the epidemiology of a communicable disease, because the threshold $R_0 = 1$ divides a self-extinguishing chain from an epidemic (Centers for Disease Control and Prevention, 2012). The same quantity can be defined for a claim: the average number of people who come to believe or reshare a piece of misinformation as a result of one believer's exposure of them. When that number exceeds one, the claim spreads exponentially; when it falls below one, the outbreak dies out. The mathematics of rumour is in fact older than the digital age — Daley and Kendall (1965) modelled rumour spread with equations adapted directly from epidemic theory half a century ago — and contemporary work has fitted formal contagion models, such as the susceptible–exposed–infected–skeptic (SEIZ) model, to the spread of specific rumours on social media with considerable accuracy (Jin et al., 2013). The epidemiologist Adam Kucharski (2020) has argued that this convergence is not coincidental: ideas and infections obey a common logic of contagion, and the reproduction number is the bridge between them.

Two dynamic features carry directly across. The first is superspreading. Just as a minority of biological cases generate the majority of transmission, a minority of accounts — the influential, the highly followed, the algorithmically favoured — generate the majority of a rumour's reach, which is why a single well-placed post can seed a national panic while a thousand ordinary ones

fade. The second is the shape of the epidemic curve itself. A claim introduced into a susceptible population with R greater than one produces the familiar wave: slow ignition, exponential rise, a peak as the susceptible pool is exhausted, and decline. Figure 2 illustrates this with a simple contagion model of a rumour under two scenarios.

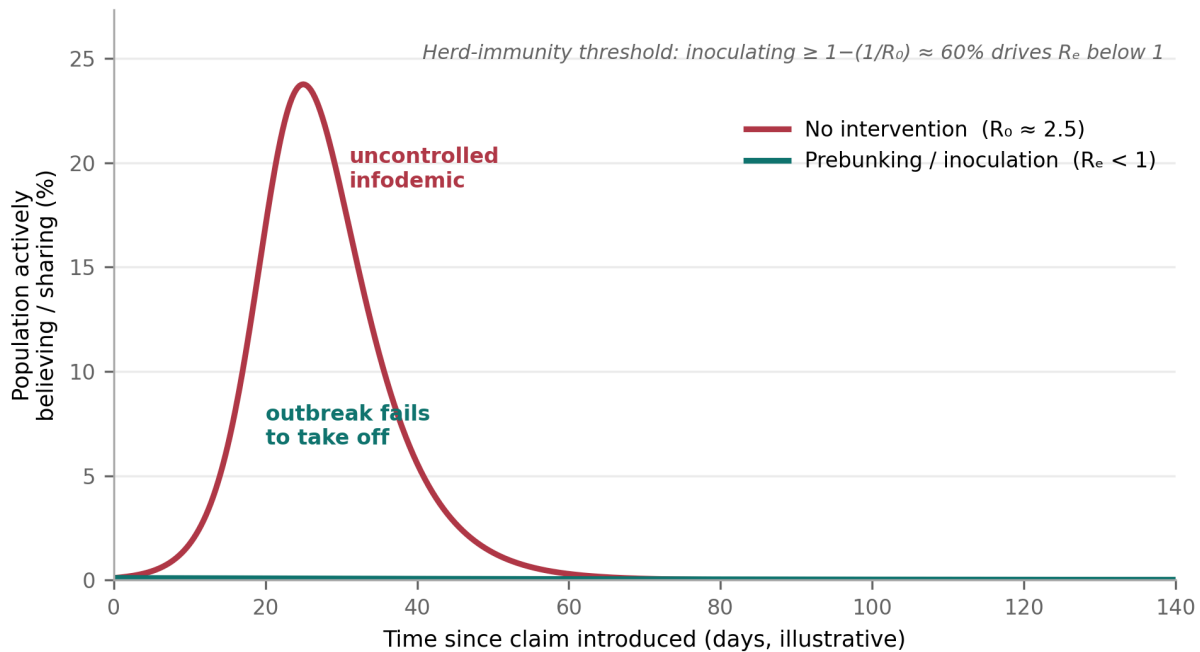


Figure 2. A contagion (SIR-type) model of rumour spread. With $R_0 \approx 2.5$ an uncontrolled infodemic sweeps the population; pre-inoculating $\geq 60\%$ (the herd-immunity threshold, $1 - 1/R_0$) pushes the effective reproduction number below 1 and the outbreak fails to take off. Parameters are illustrative.

The upper curve is an uncontrolled infodemic: with an R_0 of roughly 2.5 — a plausible value, given that false claims out-reproduce true ones — a large fraction of the population passes through the "believing" state before the wave subsides. The lower curve is the same claim in a population that has been partially inoculated in advance, so that the *effective* reproduction number, R_e , has been pushed below one; the outbreak never takes off. The gap between the two curves is the whole prize of infodemic management, and the mechanism that produces the lower curve is the subject of the next section. What the figure makes visible is that the leverage lies not in the peak but in the slope — in bending R below one before the exponential phase, exactly as in the control of a biological epidemic.

The herd-immunity threshold makes the parallel precise. In infectious-disease control, an epidemic can be prevented without immunising everyone: once the immune fraction of the population exceeds $1 - (1/R_0)$, each case produces on average fewer than one secondary case and

transmission collapses. For an R_0 of 2.5, that threshold is 60 percent. The striking finding of recent misinformation research is that an analogous threshold appears to hold for false belief. If enough of a population has been rendered resistant to a manipulation technique in advance, the technique fails to propagate even among those who were never reached directly — a protective effect that researchers have explicitly named "psychological herd immunity" (Basol et al., 2021). The reproduction number, the curve, the threshold: the quantitative core of epidemiology transfers to the infodemic not as analogy but as working model.

None of this modelling is possible without data, and here epidemiology's surveillance arm reappears as infoveillance. Just as a disease-surveillance system reads case reports, laboratory results, and syndromic signals to estimate an epidemic's trajectory in near real time, an infodemic can be tracked through the digital traces it leaves — the volume and velocity of a claim across platforms, spikes in the relevant search queries, and the sentiment and geography of its spread (Eysenbach, 2009). These signals are the case reports of the infodemic, and they allow the reproduction number of a specific rumour to be estimated while there is still time to act on it. The COVID-19 response saw the first large-scale operationalisation of such monitoring; the task now is to make it routine capacity rather than emergency improvisation.

5. Prevention: the three levels applied to the infodemic

Public health organises its response to disease into three levels of prevention — primary (stopping the disease before it occurs), secondary (detecting and treating it early), and tertiary (limiting the damage once it is established) (Centers for Disease Control and Prevention, 2012). The framework maps onto the infodemic with unusual exactness, and mapping it exposes where current responses are misallocated.

Primary prevention is inoculation, and here the analogy is not loose but literal, because the intervention was named after the medical one. Inoculation theory, developed in social psychology, holds that exposing people to a weakened form of a manipulative argument — together with a refutation of it — confers resistance to the full-strength version later, exactly as a vaccine presents a weakened antigen to train the immune system (van der Linden et al., 2017). The applied form is "prebunking": warning people about a manipulation technique before they meet it. Controlled trials of the gamified prebunking intervention *Bad News*, in which players

briefly take on the role of a misinformation producer and so learn to recognise its methods, have shown measurable and cross-culturally robust reductions in susceptibility to misinformation (Roozenbeek & van der Linden, 2019; Traber et al., 2022). This is the vaccine of the infodemic, and it is the intervention that produces the lower curve in Figure 2. Its great advantage over correction is temporal: it acts before exposure, on the slope rather than the peak.

Secondary prevention is surveillance and rapid response: detecting an emerging rumour early and treating it before it reaches the exponential phase. This is the domain of infoveillance and of the fact-checking organisation, which functions as a diagnostic and treatment service — case-finding the false claim, issuing the correction, and, ideally, doing so during the narrow window in which R can still be bent below one. Bangladesh has developed real capacity here, in independent fact-checking initiatives that monitor and debunk viral claims, and this is the layer to which the present author's own work in verification belongs. Organisations of this kind function as the sentinel surveillance and rapid-response service of the information system: they scan for the emerging claim, establish its falsity against primary sources, publish a correction, and attempt to seed that correction into the same networks the rumour is travelling. Their effectiveness is a function of speed, reach, and credibility — the epidemiological equivalents of early detection, effective case management, and a treatment the population will actually accept. But secondary prevention has an inherent handicap that the biological analogy illuminates: the correction is always slower than the infection, because it must wait for the claim to appear before it can respond, and Vosoughi and colleagues' (2018) finding that falsehood simply travels faster means the treatment perpetually chases an infection with a head start.

Tertiary prevention is harm reduction after belief is established, and it is the hardest, because misinformation exhibits a property that has no clean biological parallel: the continued-influence effect, whereby a correction fails to fully undo the damage and the false belief goes on shaping judgement even after the person accepts that it was false (Lewandowsky et al., 2012). A retracted rumour is not a cured patient; it is a chronic condition. This is precisely why the epidemiology points so strongly upstream. A disease whose treatment is weak and whose sequelae are permanent is a disease to be prevented, and the strategic implication of reading the infodemic epidemiologically is a reallocation of effort from debunking, where most resources now sit,

toward the prebunking and literacy-building that raise the population's baseline resistance before the next claim arrives.

6. The limits of the metaphor

A model is a tool, and a tool used past its domain does harm. Intellectual honesty, and the safety of the people the model is meant to serve, require marking clearly where the epidemiology of misinformation stops being illuminating and starts being dangerous — a caution that a growing critical literature has pressed with force (Simon & Camargo, 2023).

The first limit is that a pathogen is value-neutral and a claim is not. Cholera is true for everyone; there is no legitimate dispute about whether *Vibrio cholerae* should be eliminated. But "misinformation" is a judgement about content, and the boundary between falsehood and unsettled, minority, or merely inconvenient truth is neither fixed nor apolitical. Much of what was confidently labelled COVID-19 misinformation in early 2020 — about airborne transmission, about the plausibility of a laboratory origin, about the value of masks — later moved inside the boundary of legitimate debate. To model a contested claim as a pathogen is to assume the very thing in dispute, and to hand whoever controls the label the authority to declare dissent a disease.

The second limit follows, and it is the one that matters most for Bangladesh. If misinformation is an epidemic, then quarantine, suppression, and compulsory treatment present themselves as proportionate public-health measures — and applied to speech, those measures are censorship. The epidemiological frame can supply a technocratic alibi for exactly the kind of information control that a state already inclined to silence its critics will reach for. Bangladesh has lived this. The Digital Security Act of 2018, justified in part as a shield against online falsehood and rumour, became an instrument for jailing journalists and ordinary citizens for lawful criticism, its harm to public life extensively documented before it was repackaged rather than repealed. That is the iatrogenic harm of the disease model: the treatment, misapplied to a phenomenon that is only partly disease, injures the healthy tissue of public debate. An epidemiology of misinformation that does not build in this limit is not merely incomplete; it is unsafe.

The third limit is that the host is a person, not a substrate. To be "infected" with a belief is to have exercised judgement, however poorly, and the disease frame's tendency to cast the

misinformed as passive victims of contagion both misdescribes them and disrespects them. The durable interventions — inoculation, literacy, the rebuilding of institutional trust — all work precisely by treating the host as a reasoning agent to be equipped rather than a body to be sterilised. The metaphor is most useful, in the end, exactly where it remembers that it is a metaphor.

7. Conclusion: infodemiology for Bangladesh

The infodemic is not merely like an epidemic. Enough of the real structure carries across — the agent–host–environment triangle, the reproduction number and its threshold, the epidemic curve, superspreading, the three levels of prevention — that epidemiology functions as a genuine analytic framework for health misinformation and not only as a source of striking phrases. Read through that framework, the strategic conclusion is clear and it is counter-intuitive: because the treatment (correction) is slow and the sequelae (continued influence) are lasting, the decisive investment is upstream, in the primary prevention that raises a population's baseline resistance before the next virulent claim arrives.

For Bangladesh the implication is institutional. A country that has built, at real cost, a surveillance system for dengue and cholera and COVID-19 has not built the equivalent for the infodemics that run alongside them — the rumours that empty vaccination campaigns, that turned a bridge into a blood libel, that move crowds to kill. Infodemiology argues for that capacity to be made standing rather than improvised: real-time infoveillance of health rumours integrated with disease surveillance in the spirit of One Health; prebunking and media literacy treated as the routine immunisation of the information system; and independent fact-checking resourced as the emergency service it already is. But the same epidemiological seriousness that justifies this capacity also fixes its limit. The infodemic must be managed by building the population's immunity, never by the state's suppression of the claims it dislikes — because the disease model, turned into a licence for censorship, becomes the more dangerous pathogen. The right response to a bad idea, in public health as in a free society, is a better-inoculated public, not a quarantined one.

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